

## Module 2: Bootstrap Grid System

The **Bootstrap Grid System** is used to create responsive web page layouts.

Bootstrap divides the page into **12 columns**. You can combine these columns in different ways to create layouts.

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### What is Bootstrap Grid System?

The Grid System helps arrange content in rows and columns.

### Structure of Grid

Container

└─ Row

└─ Column

Example:

```
<div class="container">
  <div class="row">
    <div class="col">
      Content
    </div>
  </div>
</div>
```

---

### 1. Containers

A **Container** is the outer wrapper that holds all content.

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Bootstrap provides two types:

### **A. Fixed Width Container**

```
<div class="container">  
  Content Here  
</div>
```

Width changes according to screen size.

---

### **B. Full Width Container**

```
<div class="container-fluid">  
  Content Here  
</div>
```

Takes 100% width of the screen.

---

### **Practical Example**

```
<!DOCTYPE html>  
<html>  
<head>  
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css  
/bootstrap.min.css" rel="stylesheet">  
</head>  
<body>  
  
<div class="container bg-primary text-white p-3">  
  Fixed Width Container
```

---

</div>

<br>

<div class="container-fluid bg-success text-white p-3">

Full Width Container

</div>

</body>

</html>

---

## Output

-----  
| Fixed Width Container |  
-----

-----  
| Full Width Container |  
-----

---

## 2. Rows

A **Row** is used to create a horizontal section.

Class:

<div class="row">

</div>

Rows are placed inside containers.

### Example

```
<div class="container">  
  <div class="row">  
    Row Content  
  </div>  
</div>
```

---

### Why Rows?

Rows organize columns horizontally.

```
<div class="container">  
  <div class="row">  
  
    <div class="col bg-danger">  
      Column 1  
    </div>  
  
    <div class="col bg-warning">  
      Column 2  
    </div>  
  
  </div>  
</div>
```

---

### 3. Columns

Columns divide space into parts.

---

Bootstrap Grid contains:

12 Columns Total

---

### Example 1: Equal Columns

```
<div class="container">
  <div class="row">

    <div class="col bg-primary">
      Left
    </div>

    <div class="col bg-success">
      Right
    </div>

  </div>
</div>
```

Output:

```
-----
| Left | Right |
-----
```

Each column gets 50%.

---

### Example 2: 3 Equal Columns

```
<div class="container">
  <div class="row">

    <div class="col">One</div>
    <div class="col">Two</div>
    <div class="col">Three</div>

  </div>
</div>
```

Output:

```
-----
| One | Two | Three  |
-----
```

Each gets 33.33%.

---

## Column Sizes

Bootstrap provides:

col-1  
col-2  
col-3  
...  
col-12

---

## Example

```
<div class="container">
  <div class="row">

    <div class="col-4 bg-danger">
      4 Columns
    </div>

    <div class="col-8 bg-success">
      8 Columns
    </div>

  </div>
</div>
```

Output:

```
-----
| 4 Col | 8 Col      |
-----
```

$4 + 8 = 12$

---

### Example: 3 + 3 + 6

```
<div class="container">
  <div class="row">

    <div class="col-3 bg-primary">
      Menu
    </div>
```

```
<div class="col-3 bg-warning">
```

    Sidebar

```
</div>
```

```
<div class="col-6 bg-success">
```

    Content

```
</div>
```

```
</div>
```

```
</div>
```

---

## Practical Activity 1

Create a Two Column Layout

```
<div class="container">
```

```
  <div class="row">
```

```
    <div class="col-6 bg-primary text-white">
```

      Left Section

```
    </div>
```

```
    <div class="col-6 bg-success text-white">
```

      Right Section

```
    </div>
```

```
  </div>
```

```
</div>
```

---

## Practical Activity 2



## Create a Three Column Layout

```
<div class="container">
  <div class="row">

    <div class="col-4 bg-danger">
      Column 1
    </div>

    <div class="col-4 bg-warning">
      Column 2
    </div>

    <div class="col-4 bg-success">
      Column 3
    </div>

  </div>
</div>
```

---

## 4. Breakpoints

Breakpoints make layouts responsive.

They decide how columns behave on different screen sizes.

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### Bootstrap Breakpoints Table

#### Class Screen Size

sm    ≥ 576px

---

## Class Screen Size

md    ≥ 768px

lg     ≥ 992px

xl     ≥ 1200px

xxl    ≥ 1400px

---

## Visual Representation

Mobile



sm



md



lg



xl



xxl

---

## Example of Breakpoints

```
<div class="col-md-6">
```

Content

```
</div>
```

---

Meaning:

- Below 768px → Full Width
  - Above 768px → Half Width
- 

### Practical Example

```
<div class="container">
  <div class="row">

    <div class="col-md-6 bg-primary">
      Left
    </div>

    <div class="col-md-6 bg-success">
      Right
    </div>

  </div>
</div>
```

---

### Mobile View

```
-----
| Left  |
-----
```

```
-----
| Right |
-----
```

---

## Laptop View

-----  
| Left | Right |  
-----

---

## Multiple Breakpoints Example

```
<div class="col-sm-12 col-md-6 col-lg-4">  
  Box  
</div>
```

Meaning:

### Screen Width

Mobile 100%

Tablet 50%

Laptop 33.33%

---

## Practical Project Example

Responsive Product Layout

```
<!DOCTYPE html>  
<html>  
<head>  
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css  
/bootstrap.min.css" rel="stylesheet">  
</head>
```

---

```
<body>
```

```
<div class="container mt-5">
```

```
<div class="row">
```

```
<div class="col-sm-12 col-md-6 col-lg-4">
```

```
<div class="bg-primary text-white p-4">
```

```
Product 1
```

```
</div>
```

```
</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">
```

```
<div class="bg-success text-white p-4">
```

```
Product 2
```

```
</div>
```

```
</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">
```

```
<div class="bg-danger text-white p-4">
```

```
Product 3
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

## Interview Questions

### Q1. What is Bootstrap Grid System?

A layout system that divides a page into 12 responsive columns.

---

### Q2. What is a Container?

A wrapper that holds rows and columns.

---

### Q3. Difference between container and container-fluid?

| container   | container-fluid   |
|-------------|-------------------|
| Fixed Width | Full Width        |
| Centered    | 100% Screen Width |

---

### Q4. How many columns are available in Bootstrap Grid?

12 Columns

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### Q5. What are Breakpoints?

Breakpoints define how layouts change on different screen sizes.

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## Quick Revision

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Container



Row



Column

## Main Classes

container

container-fluid

row

col

col-1 to col-12

col-sm-\*

col-md-\*

col-lg-\*

col-xl-\*

col-xxl-\*

## Remember

Container holds content

Row holds columns

Bootstrap Grid = 12 Columns

Breakpoints create responsive layouts

Most commonly used classes:

col-md-6

col-lg-4

col-sm-12

These are heavily used in real-world websites, dashboards, admin panels, e-commerce sites, and portfolio projects.